

TEN1: Final lab report

BB2560 ADVANCED MICROBIOLOGY AND
METAGENOMICS

Summary

The sediment at the bottom of a marine environment is usually a ground for bacterial communities that break down the organic matter and act as buffer or reservoirs for future eutrophication problems (Zhang et al., 2025). The aim of this study was to comparatively analyze the aquarium sediment with the water column above. Water and sediment samples were collected from the Baltic Sea Science Centre (BSSC) and to assess the association of nitrate (NO_3^-) and phosphate (PO_4^{3-}) variability with microbial community composition. DNA was extracted using the Dneasy PowerSoil Pro Kit following the instructions of the manufacturer, amplicon libraries targeting the V3-V4 region of the 16S rRNA gene in bacteria and archaea were generated by PCR amplification. Further analysis like Alpha diversity, Beta diversity, ANOSIM/PERMANOVA was done on R Studio using the LAB 1 guide (BB2560, 2026). The water and sediment samples gave clear differences in their microbiome communities leaving space for more comparative analysis that could be done by comparing these controlled environments with real life samples from deep sea or lakes.

Introduction

A marine-like ecosystem can largely be seen in two separate layers, the Pelagic (upper water column, open sea) and the Benthic (lower sedimentary base). The bacterial communities in these layers influence the water quality and other biogeochemical cycles and could also be responsible for eutrophication (Zhong et al., 2023). The water-sediment interface is an area of interest since the microbial community here decomposes most of the organic matter that sinks to the bottom, transforming excess nutrients to maintain the ecosystem. The nutrient exchange between the sediment and overlying water directly regulates the microbial functioning (Zhang et al., 2025).

Aquariums are dynamic systems that face frequent "disturbances" like water changes, temperature shifts, or nutrient spikes. So a compositional comparison could allow us to track how the community changes over time (succession). According to (Bik et al., 2019), this method is useful for observing how artificial ecosystems' microbial communities can re-stabilize after a disturbance. Studying the changes in microbial communities could be a useful bioindicator of eutrophication and an early indication of ecosystem disturbance (Newton et al., 2011).

Therefore, the study tried to investigate the different marine-replicating ecosystems at the Baltic Sea Science Centre (BSSC) at Skansen. Three big aquariums mimicking the deep sea, open sea and coastal waters of the Baltic Sea. The aquarium walls consisted of multiple connected (partitioned) tanks mimicking deep rocky seafloors, deep soft seabeds and shallow soft seabed. There was a separate tank where only water and jellyfish were kept.

Hence, microbial community composition in water and sediment samples were analysed by extracting DNA and amplifying the 16S rRNA gene. The aim of this research report is to compare the distribution of the microbiome composition from the aquarium via 16S amplicon sequencing, as well as speculate the potential effect on the microbiome with the variability of nutrients nitrate (NO_3^-) and phosphate (PO_4^{3-}) quantities. Such comparisons may help identify microbial compositional shifts that can serve as early bioindicators of eutrophication in aquatic ecosystems in both water and sediment samples.

Materials and methods

A total of 75 sediment and aquatic samples (in total) were analysed from different marine environments of BSSC. The water was sampled with 5 L containers from the top of the tanks and were stored in the fridge at 4°C. The sediment was collected with sieves on poles, scooping up sediment from the bottom and were immediately placed on ice. Sediment samples were kept frozen at -20°C in the fridge and thawed right before the lab started. They were prepared by mixing 30g sediment with 250mL of PBS buffer. This was swirled around and then decanted onto the filters, and then filtered/treated the same way as water samples.

DNA extraction from the samples was performed using Dneasy® PowerSoil® Pro Kit (Bik et al., 2019) following the instructions of the manufacturer. Then, DNA concentration was measured using the fluorometer Qubit and following the instructions in the lab protocol, as well as the measurement of the DNA concentration (BB2560, 2026). All DNA samples with a concentration over 0.25 ng/ul were diluted to 0.25 ng/ul. The DNA was stored in -20°C (Zhong et al., 2023). Sequencing works were performed by the Illumina MiSeq sequencer (Bik et al., 2019) at SciLifeLab.

PCR was done in the Stockholm University lab for some samples while others were sent to the National Genomics Infrastructure (NGI). The 16S rRNA gene was amplified and amplicon libraries were generated by targeting the V3-V4 region of the 16S rRNA gene in bacteria and archaea using locus-specific primers 341F forward primer and 805R reverse primer (Herlemann et al., 2011) designed with adapter sequences at the 5' ends using 1X KAPA HiFi HotStart ReadyMix DNA polymerase and following the cycling conditions and Illumina 16S amplicon sequencing workflow at the National Genomics Infrastructure (NGI), SciLifeLab (*NGI Sweden » Illumina 16S Sequencing*, n.d.). DNA concentration was measured in the way already specified above.

Further bioinformatic analysis of the sequences was done using R studio. Basic packages like dada2, edgeR, pheatmap and vegan that are helpful to analyse the 16S rRNA. All samples were first analyzed, steps like sequence quality trimming, denoising, chimera removal, taxonomic annotation, normalising the ASV table, and summarising counts at different taxonomic levels were performed. Code was used from the BB2560 LAB 1 part and can be referred to in the appendix at the end of this report.

Results

Alpha Diversity analysis

Taxonomic composition study

A taxonomic composition study was done with the information obtained through the 16S rRNA gene amplicon sequencing to gain insights into the microbiome of the sediment and water samples. A higher diversity was observed at the phylum level in the sediment samples

than in the water samples (Figure 1A and 1B). The phylum Pseudomonadota and Bacteroidota dominated in the water samples (Figure 1B) while in the sediment samples (Figure 1A) Pseudomonadota, Bacteroidota and Actinomycetota were in comparable proportions.

At the family level, members of the family Flavobacteriaceae, Rhodobacteraceae and Burkholderiaceae dominated the bacterial community in the sediment samples (Figure 1C), while having a major proportion of the community “unassigned”. In the water samples, the composition of the different families was more distributed (Figure 1D) and the proportion of “unassigned” communities was negligible.

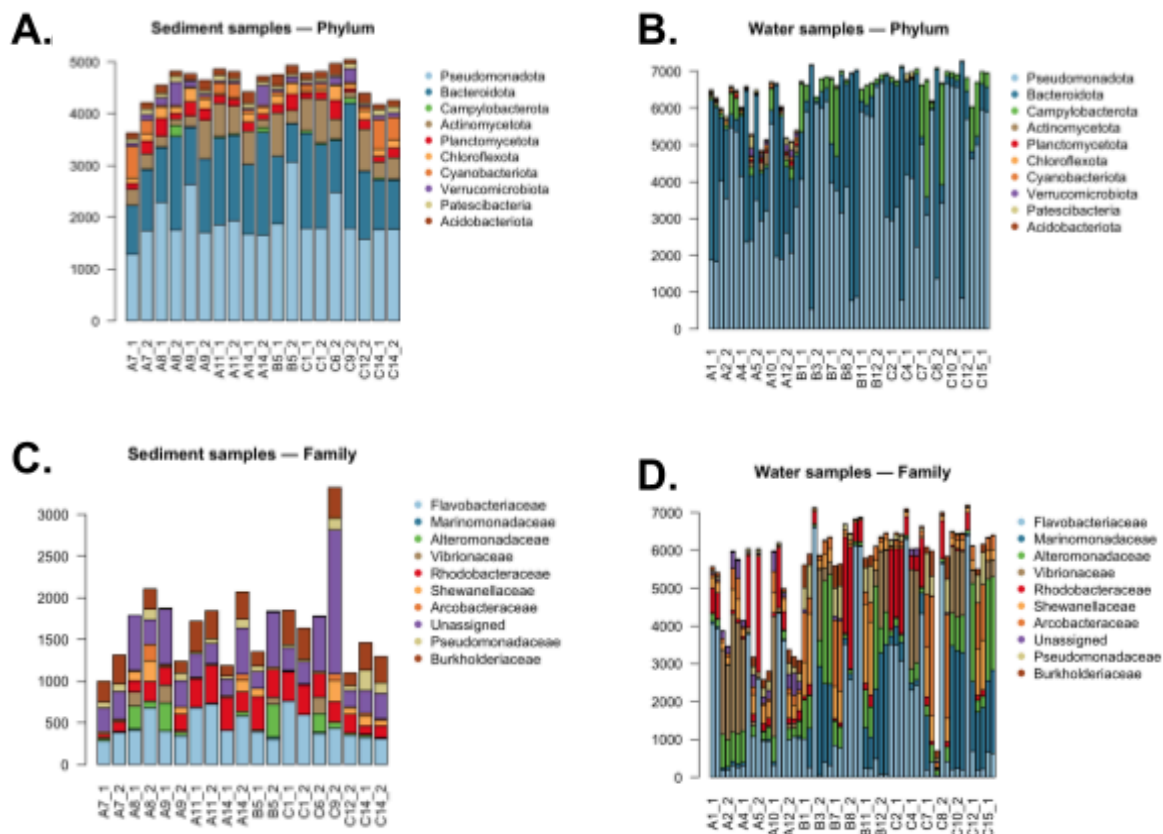


Figure 1. Microbiome composition determined by 16S rRNA gene amplicon sequencing. A. Microbiome composition of the 10 most abundant species at the phylum level of sediment samples. B. Microbiome composition of the 10 most abundant species at the phylum level of water samples. C. Microbiome composition of the 10 most abundant species at the family level of sediment samples. D. Microbiome composition of the 10 most abundant species at the family level of water samples.

Shannon diversity analysis

To study the Alpha Diversity a Shannon analysis was done on the water and the sediment samples. Wilcoxon rank sum test with continuity correction gave: $W = 10$ and a p-value of 2.114×10^{-10} , which is ($p < 0.05$) an indication for clear distinction between the water and sediment communities. Sediment samples showed a higher alpha diversity.

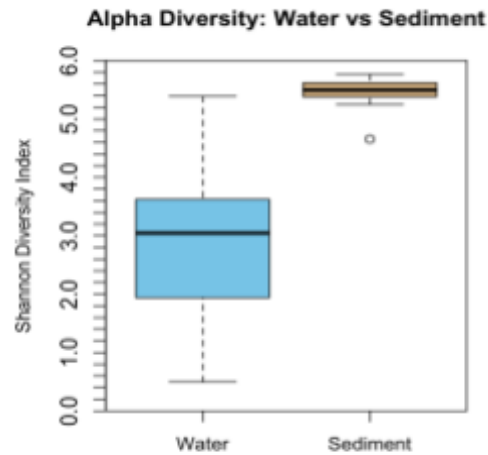
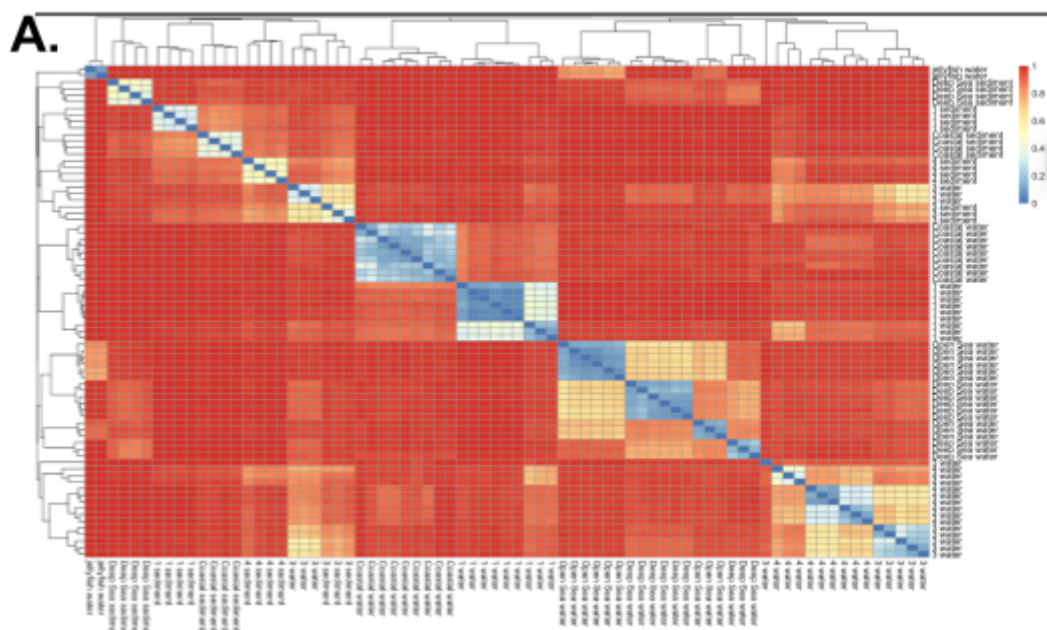


Figure 2. Boxplot summarizing Shannon diversity results for the water samples and sediment samples from BSSC that were collected. Boxes represent median, first and third quartile. Whiskers extend to the highest and lowest values that are within one and a half times the interquartile range. Sediment sample has a median of 5.49 while the water sample has a median of 3.04.

Beta-diversity Analysis

Bray-Curtis analysis

A distance matrix including pairwise community distances of the samples was calculated to study between-sample community differences. The results show a Bray-distance of 0.6-1 between the samples of water and sediment (Figure 3A) and 0.2-1 within the different sample types (Figure 3B and 3C). Hierarchical clustering based on the pairwise distances from the Bray-distance matrix was also carried out and displayed in the heatmaps, suggesting a clustering by sample type.



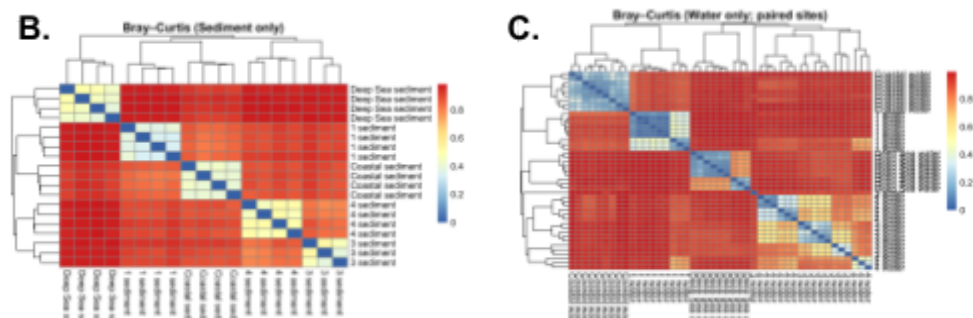


Figure 3. Hierarchical clustering beta diversity among the samples. Heatmap representing Bray-Curtis dissimilarity matrix between: A. All the water and sediment samples. B. Sediment samples only and C. Water samples only from tanks where sediment samples were also collected from.

Non-metric multidimensional scaling (NMDS) analysis

Further analysis was done by using the NMDS ordination method. It showed a clear differentiation of clustering patterns. The water samples showed 7 distinct clusters and the sediment samples showed 4 distinct clusters. Most water and sediment samples clustered largely away from each other with one cluster pair being closer (Figure 4).

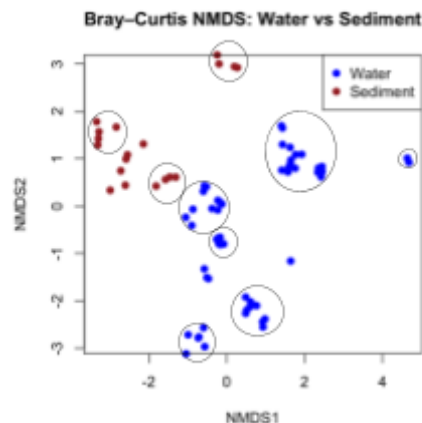


Figure 4: on-metric multidimensional scaling (NMDS) ordination beta diversity among the samples. A. All the water and sediment samples.

ANOSIM

Analysis of similarities (ANOSIM) was done to check if the clustering of samples observed was statistically significant. The results showed the following values $R = 0.5553$, $p = 0.0001$ for the water-sediment samples.

Section	Comparison	R	Significance
Water - Sediment	Water - Sediment	0.5553	1e-04

PERMANOVA

Dataset	Df	R ²	F-value	p-value
All Samples	2	0.20008	9.0047	0.0001
Water Only	1	0.15811	10.141	0.0001
Sediment Only	1	0.22561	4.9529	0.0001

Table 1: Permutational multivariate analysis of variance (PERMANOVA) between samples. Sample type (All samples), Sediment samples and Water samples from which sediment samples were also collected.

To gain deeper insights into the impact of nutrients in the environment ($\text{NO}_3^- + \text{PO}_4^{3-}$) on the microbial composition, a Permutational Multivariate Analysis of Variance (PERMANOVA) was done. Across all samples, nutrient concentration significantly influenced microbial beta diversity ($R^2 = 0.200$, $F = 9.00$, $p = 0.0001$), indicating that approximately 20% of the observed variation in community composition could be explained by combined nitrate and phosphate levels. Even in the water and sediment habitat separately, this nutrient concentration showed to be a significant factor (water ($R^2 = 0.158$, $F = 10.14$, $p = 0.0001$) and sediment ($R^2 = 0.226$, $F = 4.95$, $p = 0.0001$)).

Discussion

This study aimed to compare microbial community composition between pelagic (water column) and benthic (sediment) environments within a controlled aquatic system. Based on previous studies (Zhang et al., 2025), sediment samples were expected to have more alpha diversity and taxonomic composition as compared to the water samples.

Alpha Diversity Analysis

Due to the difficulties to access deep-sea habitats, there is only a limited knowledge about the deep-sea benthic microbiome and hence we see a large proportion of “unassigned” species in the compositional analysis (Barrenechea Angeles et al., 2024).

Pelagic and benthic bacterial communities were distinctly different in terms of diversity, taxonomic composition and community structure, with Pseudomonadota, Bacteroidota, and Campylobacterota dominating the water, and Pseudomonadota, Bacteroidota, Actinomycetota (Bik et al., 2019), Chloroflexota and Cyanobacteriota (Zhang et al., 2025) and Plantomycetota (Barrenechea Angeles et al., 2024) dominating sediment potentially reflecting their metabolic capacity for anaerobic respiration and participation nitrogen cycling processes under reduced oxygen conditions.

Beta Diversity Analysis

Both analyses (Bray-Curtis and NMDS) demonstrated clear clustering by sample type (water vs. sediment), indicating distinct microbial communities in these environments. There was

only one pair of clusters from both samples that was closer which could be the different species from Bacteroidota phylum, this is

ANOSIM and PERMANOVA

ANOSIM results indicated that the habitat plays an important role in structuring microbial communities within the aquarium system. In Permanova, The higher R^2 observed (Anderson, 2001) in the sediment samples suggests that benthic microbial assemblages may be more strongly influenced by nutrient accumulation compared to the water communities. These are consistent with previous studies where environmental variables like ammonia were shown to significantly impact the composition (Oliaro et al., 2025).

Limitations and Future directions

The relatively high proportions of unassigned families in the sediment samples is reflective of the limitations about difficulty in access to real world sea beds mentioned in (Barrenechea Angeles et al., 2024). This could be a potential hidden ground for further analyses on ARGs or other organisms that play a crucial role in maintaining the ecology of the marine environments. Instead of only NO_3^- and PO_4^{3-} we could also see the dependence of these communities on other factors like the salinity, temperature, pH. The 'still-low-flow' nature of the aquarium may not be able to mimic the natural aquatic environment so comparing samples from the real world to these could help understand the community composition better and help identifying which species might not be surviving in the aquariums. We can include metagenomic analysis to controlled nutrient fluctuations (changes in $\text{NO}_3^- + \text{PO}_4^{3-}$) environments to find functional genes without needing a reference and hence using the compositional analysis as a predictive indicator for eutrophication.

Word count

Figures and captions not included.

Summary: ; Introduction: 331; Materials and methods: 348; Results: 467; Discussion:440

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APPENDIX: Bioinformatic Analysis Pipeline (R Code)

R code used in reference with the LAB1 tutorial in the BB2560, 2026 course and self coded parts.

```
#Changes fro TEN1 from LAB1
# sample_name -> course_id
# changed code for clustering and reading fnFs and fnRs
# seqtab -> seqtab

# To check chimeras across all the samples created a dataframe with sample_id and chimeric count columns

##! nonchim_df <- data.frame(
#   Sample = colnames(seqtab),
#   NonChimeric_ASVs = colSums(seqtab > 0)
# )

# nonchim_df

library(dada2)
library(edgeR)
library(pheatmap)
library(vegan)

sample_info = read.delim("sample_info.txt")
sample_info

ngi_sample_id = sample_info[,1]
course_id = sample_info[,2]
sample_label = sample_info[,3]
sample_temperature = sample_info[,4]
sample_salinity = sample_info[,5]
sample_pH = sample_info[,6]
sample_PO4_levels = sample_info[,7]
sample_NO3_levels = sample_info[,8]

sample_PO4_levels
sample_NO3_levels
```

```

# Find position in the list of forward read filenames that matches each NGI ID
ix = sapply(
  ngi_sample_id, grep,
  list.files(pattern="_R1.fastq")
)

# ...then extract the filenames of forward reads
r1_files <- list.files(
  recursive = TRUE,
  pattern = "_R1.fastq",
  full.names = TRUE
)

ix <- sapply(ngi_sample_id, function(id)
  grep(id, r1_files)
)

fnFs <- r1_files[ix]
#fnFs = list.files(pattern="_R1.fastq")[ix]

# Find position in the list of reverse read filenames that matches each NGI ID
ix = sapply(
  ngi_sample_id, grep,
  list.files(pattern="_R2.fastq")
)

# ...then extract the filenames of reverse reads
#fnRs = list.files(pattern="_R2.fastq")[ix]
r2_files <- list.files(
  recursive = TRUE,
  pattern = "_R2.fastq",
  full.names = TRUE
)

ix <- sapply(ngi_sample_id, function(id)
  grep(id, r2_files)
)

fnRs <- r2_files[ix]

# Check that the forward and reverse fastq files come in the same order
fnFs
fnRs

plotQualityProfile(fnFs[1:4])
plotQualityProfile(fnRs[1:4])

filtFs = file.path("filtered", paste0(ngi_sample_id, "_F_filt.fastq.gz"))
filtRs = file.path("filtered", paste0(ngi_sample_id, "_R_filt.fastq.gz"))

#out = filterAndTrim(
#  fnFs, filtFs, fnRs, filtRs,
#  truncLen=c(180,180), trimLeft=c(19,21),
#  maxN=0, maxEE=c(2,2), truncQ=2, rm.phix=TRUE,
#  compress=TRUE, multithread=TRUE
#)
out = filterAndTrim(
  fnFs, filtFs, fnRs, filtRs,
  truncLen=c(220,220),
  maxN=0, maxEE=c(2,2), truncQ=2, rm.phix=TRUE,
  compress=TRUE, multithread=TRUE
)
out

# For a few of forward files
plotQualityProfile(filtFs[1:4])

# And a few of reverse files
plotQualityProfile(filtRs[1:4])

# we start with the forward reads
errF = learnErrors(filtFs, multithread=TRUE)

# and the same for the reverse reads
errR = learnErrors(filtRs, multithread=TRUE)

```

```

# Run dereplication
derepFs = derepFastq(filtFs, verbose=TRUE)
derepRs = derepFastq(filtRs, verbose=TRUE)

# Name the derep-class objects by the sample names:
names(derepFs) = course_id
names(derepRs) = course_id

derepFs
derepRs

dadaFs = dada(derepFs, err=errF, multithread=TRUE)
dadaRs = dada(derepRs, err=errR, multithread=TRUE)

dadaFs
dadaRs

mergers = mergePairs(dadaFs, derepFs, dadaRs, derepRs, verbose=TRUE)
seqtab = makeSequenceTable(mergers)

#seqtab = removeBimeraDenovo(
# seqtab, method="consensus", multithread=TRUE, verbose=TRUE

seqtab = removeBimeraDenovo(
  seqtab, method="consensus", multithread=TRUE, verbose=TRUE
)

#seqtab = t(seqtab)
seqtab = t(seqtab)
# to check the column names
colnames(seqtab)

# to check the row names
rownames(seqtab)

# to copy the ASV sequences into a new vector
asv = rownames(seqtab)

# to rename the rows
rownames(seqtab) = paste("ASV", 1:length(asv), sep = "")

# to check the new row names
rownames(seqtab)

# Chimeric analysis
#The samples are printed in the same order as they appear in the seqtab columns (and in sample_name).

# Create a dataframe that stores the number of non-chimeric ASVs detected in each sample.
# For every sample (column in seqtab), this counts how many ASVs have a non-zero abundance (>0),
# i.e., the number of ASVs present in that sample after chimera removal.

nonchim_df <- data.frame(
  Sample = colnames(seqtab),
  NonChimeric_ASVs = colSums(seqtab > 0)
)

nonchim_df

#view seqtab
View(seqtab)
#view nonchim data frame
View(nonchim_df)

#-----Ecology and Taxonomy-----#

# set the seed
set.seed(100)

taxa_genus = assignTaxonomy(
  asv, "sbdi-gtdb.assignTaxonomy_no_species_or_kingdom.fna.gz",
  multithread=TRUE,
  taxLevels = c(
    "Domain", "Phylum", "Class", "Order", "Family", "Genus"
  )
)

```

```

)
)

taxa = addSpecies(
  taxa_genus,
  "sbdi-gtdb.20genomes.addSpecies.fna.gz",
  allowMultiple = FALSE,
  tryRC = FALSE,
  n = 2000,
  verbose = FALSE
)

# we name the rows of taxa as the rows of seqtab, ie with ASV ids
rownames(taxa) = rownames(seqtab)

# check what's in the different columns
colnames(taxa)

# to for example see the class-level annotation of the 3rd ASV, do:
taxa[3,"Class"]

# to see the species of the 111st ASV, do:
taxa[111,"Species"]

ix = which(!is.na(taxa[, 'Species']))
taxa[ix, 'Species'] = paste(taxa[ix, 'Genus'], taxa[ix, 'Species'])

colSums(!is.na(taxa))
View(taxa)
View(seqtab) # Look for high counts
asv[2] # Get sequence of ASVs
asv[3] # Get sequence of ASVs
asv[8] # Get sequence of ASVs
asv[14] # Get sequence of ASVs
asv[600] # Get sequence of ASVs

norm_seqtab = seqtab
for (i in 1:ncol(seqtab)) {
  norm_seqtab[,i] = seqtab[,i]/sum(seqtab[,i])
}

clade_counts = list()
norm_clade_counts = list()

for (i in 1:ncol(taxa)) {
  matr = norm_matr = NULL
  clade = unique(taxa[,i])
  clade = clade[!is.na(clade)]
  for (j in 1:length(clade)) {
    ix = which(clade[j]==taxa[,i])
    if (length(ix) > 1) {
      matr = rbind(matr, apply(seqtab[ix,], 2, sum, na.rm=TRUE))
      norm_matr = rbind(norm_matr, apply(norm_seqtab[ix,], 2, sum, na.rm=TRUE))
    } else {
      matr = rbind(matr, seqtab[ix,])
      norm_matr = rbind(norm_matr, norm_seqtab[ix,])
    }
  }
  rownames(matr) = rownames(norm_matr) = clade
  colnames(matr) = colnames(norm_matr) = course_id
  clade_counts[[i]] = matr
  norm_clade_counts[[i]] = norm_matr
}

#check sample salinity
sample_salinity

# Define the names for each list element
taxonomic_levels <- c("Domain", "Phylum", "Class", "Order", "Family", "Genus", "Species")

# Assign these names to the list
names(clade_counts) <- taxonomic_levels
names(norm_clade_counts) <- taxonomic_levels

```

```

# Let's check the counts of all phyla in the first sample. Phylum is the third level in the taxonomy.
clade_counts[["Phylum"]][,75]

# Or the counts of the third genus in samples 1-10. The genus is the seventh level.
clade_counts[["Genus"]][3,1:75]

make_taxonomic_bargraph <- function(clade_abundance, sample_names, tax_level, cex_names = 1, cex_taxa = -1,
main = "", top_x = -1, sorting = NULL) {

  # Set a default sorting order
  if (is.null(sorting)) {sorting = 1:ncol(clade_abundance[[tax_level]])}

  # Select which clades to plot
  if (top_x != -1) {
    ok = sort(rowMeans(clade_abundance[[tax_level]]), index.return = T, decreasing = T)$ix[1:top_x]
  } else {
    ok = 1:nrow(clade_abundance[[tax_level]])
  }

  # Set color palette
  mycols =
colorRampPalette(c("#A6CEE3", "#1F78B4", "#B2DF8A", "#33A02C", "#FB9A99", "#E31A1C", "#FDBF6F", "#FF7F00", "#CAB2D6", "#6A3D9A", "#FFFF99", "#B15928"))

  # Adjust margins to make space for the legend
  old_par <- par(mar = c(5, 4, 4, 2), oma = c(0, 0, 0, 10)) # reserve outer space on right # Increase the
right margin

  barplot(clade_abundance[[tax_level]][ok,sorting],
          col = mycols(length(ok)),
          las = 2,
          cex.names = cex_names,
          names.arg = sample_names[sorting],
          main = main)

  # calculate the size of the taxa names
  if (cex_taxa == -1) {cex_taxa = min(1.2, 10/length(ok))}

  # Add the legend to the right of the plot
  # calculate the size of the taxa names
  if (cex_taxa == -1) {cex_taxa = min(2, 10/length(ok))}

  # Draw legend in the right outer margin (no overlap)
  par(xpd = NA)
  legend(
    x = par("usr")[2] + 0.5, # just right of plotting region
    y = par("usr")[4], # near the top
    legend = rownames(clade_abundance[[tax_level]][ok]),
    col = mycols(length(ok)),
    pch = 19,
    bty = "n",
    cex = cex_taxa,
    ncol = 1 # number of oclumns for legend
  )
  # Restore original margins
  par(old_par)
}

# to make water vs sediment indices from sample_label
# Use the Label column to classify sample type
is_water <- grepl("\\bwater\\b", sample_label, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", sample_label, ignore.case = TRUE)

table(is_water, is_sediment)
#is_water
#is_sediment

# 0) Check objects exist
exists("seqtab")
exists("taxa")

# 1) Align taxa rows to seqtab rows (ASV IDs)
stopifnot(all(rownames(seqtab) %in% rownames(taxa)))
taxa2 <- taxa[rownames(seqtab), , drop = FALSE]

```

```

# 2) Build clade abundance (taxon x samples) for each tax level
make_clade_abundance_ASVrows <- function(seqtab_asv_by_sample, taxa,
                                         levels = c("Domain", "Phylum", "Class", "Order", "Family", "Genus")) {
  out <- list()
  for (lvl in levels) {
    grp <- taxa[, lvl]
    grp[is.na(grp) | grp == ""] <- "Unassigned"
    out[[lvl]] <- rowsum(seqtab_asv_by_sample, group = grp) # taxon x samples
  }
  out
}

# 3) CREATE clade abundance (this is the key line)
clade_abundance <- make_clade_abundance_ASVrows(seqtab, taxa2)

# 4) Verify
names(clade_abundance)
dim(clade_abundance[["Genus"]])

tax_level <- "Genus"
ab <- clade_abundance[[tax_level]]

# reorder metadata to match sample columns
ord <- match(colnames(ab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]

is_water_ab <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment_ab <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)

sorting_wat <- which(is_water_ab)
sorting_sed <- which(is_sediment_ab)

#Phylum Level barplots
tax_level <- "Phylum" # or "Phylum" for a cleaner figure
ab <- clade_abundance[[tax_level]]

make_taxonomic_bargraph(
  clade_abundance = clade_abundance,
  sample_names = colnames(ab),
  tax_level = tax_level,
  sorting = sorting_wat,
  top_x = 10,
  main = paste0("Water samples - ", tax_level)
)

make_taxonomic_bargraph(
  clade_abundance = clade_abundance,
  sample_names = colnames(ab),
  tax_level = tax_level,
  sorting = sorting_sed,
  top_x = 10,
  main = paste0("Sediment samples - ", tax_level)
)

names(clade_abundance)

#family level bargraph
tax_level <- "Family"
ab <- clade_abundance[[tax_level]]

# (Optional) choose water/sediment sorting like before
ord <- match(colnames(ab), course_id)
stopifnot(!any(is.na(ord)))
label_ab <- sample_label[ord]
is_water_ab <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment_ab <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)

sorting_wat <- which(is_water_ab)

```

```

sorting_sed <- which(is_sediment_ab)

make_taxonomic_bargraph(
  clade_abundance = clade_abundance,
  sample_names = colnames(ab),
  tax_level = tax_level,
  sorting = sorting_wat,
  top_x = 10,
  main = "Water samples - Family"
)

make_taxonomic_bargraph(
  clade_abundance = clade_abundance,
  sample_names = colnames(ab),
  tax_level = tax_level,
  sorting = sorting_sed,
  top_x = 10,
  main = "Sediment samples - Family"
)

make_taxonomic_bargraph(clade_abundance = norm_clade_counts, sample_names = paste(sample_label, course_id),
tax_level = "Class", top_x = 8, cex_names = 0.6, main = "Relative abundance of the 8 most abundant classes")

# SHANON INDEX for ALPHA DIVERSITY
# First we load the vegan package
library(vegan)

# Then calculate Shannon diversity for every sample and put in a vector named shannon
shannon = diversity(seqtab, MARGIN = 2)

# You MUST align metadata to sample order (same step we did for barplots):
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]
# We can make a bargraph of the Shannon diversities.
# - We order the samples using the "all" group
# identify the water vs sediment samples
is_water <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)

# barplot(
#   shannon[all], las = 2,
#   names.arg = paste(sample_type[all], sample_name[all])
# )

barplot(
  shannon,
  las = 2,
  names.arg = label_ab,
  cex.names = 0.6,
  main = "Shannon Diversity Index per Sample"
)

boxplot(
  shannon[is_water],
  shannon[is_sediment],
  names = c("Water", "Sediment"),
  col = c("skyblue", "tan"),
  ylab = "Shannon Diversity Index",
  yaxt = "n",
  ylim = c(0,6),
  main = "Alpha Diversity: Water vs Sediment"
)

axis(2, at = seq(0,6,0.2), cex.axis = 1.2)

wilcox.test(
  shannon[is_water],
  shannon[is_sediment]
)

median(shannon[is_sediment])

```

```

median(shannon[is_water])

# Beta-Diversity
#Bray Curtis
library(vegan)
library(pheatmap)

# Ensure metadata aligns with sample columns in seqtab
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]
NO3_ab <- as.numeric(sample_NO3_levels[ord])
PO4_ab <- as.numeric(sample_PO4_levels[ord])

# Define habitat groups
is_water <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)
habitat <- ifelse(is_water, "Water", ifelse(is_sediment, "Sediment", "Other"))

# Normalize to relative abundance per sample (recommended before Bray-Curtis)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

bray_dist <- as.matrix(vegdist(t(norm_seqtab), method = "bray"))

# Look at it (optional)
bray_dist[1:5, 1:5]

# Heatmap without clustering
pheatmap(
  bray_dist,
  cluster_rows = FALSE, cluster_cols = FALSE,
  labels_row = label_ab,
  labels_col = label_ab
)

# Heatmap WITH clustering (recommended)
pheatmap(
  bray_dist,
  clustering_distance_rows = as.dist(bray_dist),
  clustering_distance_cols = as.dist(bray_dist),
  labels_row = label_ab,
  labels_col = label_ab
)

set.seed(100)
mds <- metaMDS(as.dist(bray_dist), k = 2, trymax = 100)

color_type <- rep("grey", length(habitat))
color_type[habitat == "Sediment"] <- "brown"
color_type[habitat == "Water"] <- "blue"

# Use a single point shape since you don't have cultured/uncultured
pch_type <- rep(21, length(habitat)) # filled circle

par(mar=c(5,4,4,8), xpd = TRUE)
plot(
  mds$points[,1], mds$points[,2],
  cex = 2, pch = pch_type,
  col = "black", bg = color_type,
  xlab = "NMDS1", ylab = "NMDS2",
  main = "NMDS (Bray-Curtis): Water vs Sediment"
)
legend("topright", legend = c("Water", "Sediment"),
  pt.bg = c("blue", "brown"), pch = 21, bty = "n", inset = c(-0.15, 0))

these <- which(habitat %in% c("Water", "Sediment"))
anosim(as.dist(bray_dist[these, these]), habitat[these], permutations = 9999)

eutro <- NO3_ab + PO4_ab
adonis2(as.dist(bray_dist) ~ eutro + habitat, permutations = 9999)

```

```

library(vegan)

pairwise_anosim <- function(bray_mat, group_vec, g1, g2, perms = 9999) {
  these <- which(group_vec %in% c(g1, g2))
  stopifnot(length(these) > 2)

  a <- anosim(as.dist(bray_mat[these, these]),
              grouping = group_vec[these],
              permutations = perms)

  data.frame(
    Comparison = paste(g1, "-", g2),
    R = unname(a$statistic),
    Significance = a$signif,
    stringsAsFactors = FALSE
  )
}

# Ensure habitat is exactly "Water"/"Sediment"
habitat2 <- factor(habitat, levels = c("Water", "Sediment"))

tab_A <- pairwise_anosim(bray_dist, habitat2, "Water", "Sediment", perms = 9999)
tab_A

install.packages("gridExtra")
library(grid)
library(gridExtra)

# Combine sections like the example layout
tab_A$Section <- "Water - Sediment"

# If you created tab_water/tab_sed, add them too:
# tab_water$Section <- "Water (between tanks)"
# tab_sed$Section <- "Sediment (between tanks)"

final_tab <- tab_A # or rbind(tab_A, tab_water, tab_sed)

# Nice formatting (scientific p-values)
final_tab$Significance <- format(final_tab$Significance, scientific = TRUE, digits = 2)
final_tab$R <- round(final_tab$R, 4)

# Build table grob
tg <- tableGrob(final_tab[, c("Section", "Comparison", "R", "Significance")],
  rows = NULL)

grid.newpage()
grid.arrange(
  textGrob("Table. Analysis of similarities (ANOSIM) between sample groups",
    gp = gpar(fontsize = 16, fontface = "bold")),
  tg,
  ncol = 1,
  heights = c(0.15, 0.85)
)

library(vegan)

# indices for each habitat
ix_wat <- which(habitat == "Water")
ix_sed <- which(habitat == "Sediment")

# Bray-Curtis within each habitat (remember: vegdist needs samples as rows)
bray_wat <- vegdist(t(norm_seqtan)[ix_wat, , drop=FALSE], method = "bray")
bray_sed <- vegdist(t(norm_seqtan)[ix_sed, , drop=FALSE], method = "bray")

set.seed(100)
nmDS_wat <- metaMDS(bray_wat, k = 2, trymax = 100)

set.seed(100)
nmDS_sed <- metaMDS(bray_sed, k = 2, trymax = 100)

plot(nmDS_wat$points,
  pch = 19, cex = 1.5,
  xlab = "NMDS1", ylab = "NMDS2",

```

```

    main = paste0("Water only (Bray-Curtis NMDS), stress=", round(nmds_wat$stress, 3))
# text(nmds_wat$points, labels = label_ab[ix_wat], cex = 0.6, pos = 3)

plot(nmds_sed$points,
     pch = 19, cex = 1.5,
     xlab = "NMDS1", ylab = "NMDS2",
     main = paste0("Sediment only (Bray-Curtis NMDS), stress=", round(nmds_sed$stress, 3)))
#text(nmds_sed$points, labels = label_ab[ix_sed], cex = 0.6, pos = 3)

# --- Beta diversity: separate NMDS for Water and Sediment, colored by eutrophication (NO3+PO4) ---

library(vegan)

# 0) Align metadata to seqtab sample columns (seqtab is ASVs x samples)
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]
NO3_ab <- as.numeric(sample_NO3_levels[ord])
PO4_ab <- as.numeric(sample_PO4_levels[ord])

# 1) Define habitat groups from sample labels
habitat <- ifelse(grepl("\\bwater\\b", label_ab, ignore.case = TRUE), "Water",
                  ifelse(grepl("\\bsediment\\b", label_ab, ignore.case = TRUE), "Sediment", "Other"))

ix_wat <- which(habitat == "Water")
ix_sed <- which(habitat == "Sediment")
stopifnot(length(ix_wat) > 2, length(ix_sed) > 2)

# 2) Normalize counts to relative abundance per sample (recommended for Bray-Curtis)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

# 3) Eutrophication index + robust grouping
eutro <- NO3_ab + PO4_ab

# Try 3 bins (Low/Mid/High). If quantile breaks are not unique, fall back to Low/High.
cuts <- unique(quantile(eutro, probs = c(0, 1/3, 2/3, 1), na.rm = TRUE))

if (length(cuts) >= 4) {
  eutro_group <- cut(eutro, breaks = cuts, include.lowest = TRUE,
                    labels = c("Low", "Mid", "High"))
  cols <- c("Low" = "skyblue", "Mid" = "orange", "High" = "red")
} else {
  eutro_group <- ifelse(eutro <= median(eutro, na.rm = TRUE), "Low", "High")
  eutro_group <- factor(eutro_group, levels = c("Low", "High"))
  cols <- c("Low" = "skyblue", "High" = "red")
}

# 4) Bray-Curtis distances within each habitat (vegdist needs samples as rows)
bray_wat <- vegdist(t(norm_seqtab)[ix_wat, , drop = FALSE], method = "bray")
bray_sed <- vegdist(t(norm_seqtab)[ix_sed, , drop = FALSE], method = "bray")

# 5) NMDS within each habitat
set.seed(100)
nmds_wat <- metaMDS(bray_wat, k = 2, trymax = 100)

set.seed(100)
nmds_sed <- metaMDS(bray_sed, k = 2, trymax = 100)

# 6) Plot side-by-side
par(mfrow = c(1, 2), mar = c(5, 4, 4, 2))

plot(nmds_wat$points,
     pch = 19, cex = 1.5,
     col = cols[as.character(eutro_group[ix_wat])],
     xlab = "NMDS1", ylab = "NMDS2",
     main = paste0("Water only (Bray-Curtis)\nStress = ", round(nmds_wat$stress, 3)))
legend("topright", legend = names(cols), col = cols, pch = 19, bty = "n")

plot(nmds_sed$points,
     pch = 19, cex = 1.5,
     col = cols[as.character(eutro_group[ix_sed])],
     xlab = "NMDS1", ylab = "NMDS2",
     main = paste0("Sediment only (Bray-Curtis)\nStress = ", round(nmds_sed$stress, 3)))
legend("topright", legend = names(cols), col = cols, pch = 19, bty = "n")

```

```

par(mfrow = c(1, 1))

# 7) Optional: PERMANOVA within each habitat testing eutrophication effect (continuous)
adonis2(bray_wat ~ eutro[ix_wat], permutations = 9999)
adonis2(bray_sed ~ eutro[ix_sed], permutations = 9999)

# --- Bray-Curtis (within-habitat) for Water-only and Sediment-only ---

library(vegan)
library(pheatmap)

# 0) Align metadata to seqtab sample columns (seqtab is ASVs x samples)
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]

# 1) Define habitat (Water/Sediment) from labels
habitat <- ifelse(grepl("\\bwater\\b", label_ab, ignore.case = TRUE), "Water",
  ifelse(grepl("\\bsediment\\b", label_ab, ignore.case = TRUE), "Sediment", "Other"))

ix_wat <- which(habitat == "Water")
ix_sed <- which(habitat == "Sediment")
stopifnot(length(ix_wat) > 2, length(ix_sed) > 2)

# 2) Normalize counts to relative abundance per sample (recommended for Bray-Curtis)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

# 3) Bray-Curtis distances WITHIN water samples only
bray_wat <- as.matrix(vegdist(t(norm_seqtab)[ix_wat, , drop = FALSE], method = "bray"))

# 4) Bray-Curtis distances WITHIN sediment samples only
bray_sed <- as.matrix(vegdist(t(norm_seqtab)[ix_sed, , drop = FALSE], method = "bray"))

# 5) Optional: view matrices
bray_wat[1:5, 1:5]
bray_sed[1:5, 1:5]

# 6) Heatmaps (within-habitat only)
pheatmap(bray_wat,
  clustering_distance_rows = as.dist(bray_wat),
  clustering_distance_cols = as.dist(bray_wat),
  main = "Bray-Curtis (Water only)",
  labels_row = label_ab[ix_wat],
  labels_col = label_ab[ix_wat])

pheatmap(bray_sed,
  clustering_distance_rows = as.dist(bray_sed),
  clustering_distance_cols = as.dist(bray_sed),
  main = "Bray-Curtis (Sediment only)",
  labels_row = label_ab[ix_sed],
  labels_col = label_ab[ix_sed])

library(vegan)
library(pheatmap)

# 1) Align metadata to seqtab
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]
cid_ab <- course_id[ord]

# 2) Habitat classification
is_water <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)

# 3) Extract tank prefix (e.g. A1 from A1_1)
tank_prefix <- sub("_.*", "", cid_ab)

# 4) Tanks that have BOTH water and sediment
tanks_wat <- unique(tank_prefix[is_water])

```

```

tanks_sed <- unique(tank_prefix[is_sediment])

paired_tanks <- intersect(tanks_wat, tanks_sed)

# 5) Keep ONLY paired WATER samples
ix_wat_paired <- which(is_water & tank_prefix %in% paired_tanks)

# 6) Normalize seqtab (relative abundance)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

# 7) Bray-Curtis within paired WATER samples
bray_wat_paired <- as.matrix(
  vegdist(t(norm_seqtab)[ix_wat_paired, , drop=FALSE], method = "bray")
)

library(vegan)
library(pheatmap)

# 0) Align metadata to seqtab columns
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- as.character(sample_label[ord])

# 1) Identify habitat from sample_label
is_water <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)

# 2) Build a "site key" that is the same for water + sediment counterparts
# Examples:
# "Deep Sea water" -> "deep sea"
# "3 sediment" -> "3"
# "Coastal sediment" -> "coastal"
site_key <- tolower(label_ab)
site_key <- gsub("\\bwater\\b|\\bsediment\\b", "", site_key)
site_key <- gsub("\\s+", " ", trimws(site_key))

# 3) Find which sites have BOTH water and sediment
sites_with_sed <- unique(site_key[is_sediment])
sites_with_wat <- unique(site_key[is_water])
paired_sites <- intersect(sites_with_wat, sites_with_sed)

# 4) Keep ONLY water samples from paired sites
ix_wat_paired <- which(is_water & site_key %in% paired_sites)

# (optional sanity check)
cat("Water samples kept:", length(ix_wat_paired), "\n")
cat("Excluded water sites (no sediment counterpart):",
  setdiff(sites_with_wat, paired_sites), "\n")

# 5) Normalize counts to relative abundance per sample (recommended for Bray-Curtis)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

# 6) Bray-Curtis among the paired water samples only
bray_wat_paired <- as.matrix(
  vegdist(t(norm_seqtab)[ix_wat_paired, , drop = FALSE], method = "bray")
)

# 7) Heatmap
pheatmap(
  bray_wat_paired,
  clustering_distance_rows = as.dist(bray_wat_paired),
  clustering_distance_cols = as.dist(bray_wat_paired),
  labels_row = label_ab[ix_wat_paired],
  labels_col = label_ab[ix_wat_paired],
  main = "Bray-Curtis (Water only; only sites with sediment counterpart)"
)

library(vegan)
library(pheatmap)

# --- Bray-Curtis within WATER-only and within SEDIMENT-only,
# keeping ONLY sites that have BOTH water + sediment counterparts ---

# 0) Align metadata to seqtab columns

```

```

ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- as.character(sample_label[ord])

# 1) Identify habitat from sample_label
is_water <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)

# 2) Build site key shared by water/sediment (remove the habitat word)
site_key <- tolower(label_ab)
site_key <- gsub("\\bwater\\b|\\bsediment\\b", "", site_key)
site_key <- gsub("\\s+", " ", trimws(site_key))

# 3) Sites that have BOTH water and sediment
paired_sites <- intersect(unique(site_key[is_water]), unique(site_key[is_sediment]))

# 4) Indices for paired water and paired sediment samples
ix_wat_paired <- which(is_water & site_key %in% paired_sites)
ix_sed_paired <- which(is_sediment & site_key %in% paired_sites)

cat("Paired sites:", paired_sites, "\n")
cat("Water kept:", length(ix_wat_paired), " Sediment kept:", length(ix_sed_paired), "\n")

# 5) Normalize to relative abundance per sample (recommended)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

# 6) Bray-Curtis within paired WATER samples only
bray_wat_paired <- as.matrix(
  vegdist(t(norm_seqtab)[ix_wat_paired, , drop = FALSE], method = "bray")
)

# 7) Bray-Curtis within paired SEDIMENT samples only
bray_sed_paired <- as.matrix(
  vegdist(t(norm_seqtab)[ix_sed_paired, , drop = FALSE], method = "bray")
)

# 8) Heatmaps
pheatmap(
  bray_wat_paired,
  clustering_distance_rows = as.dist(bray_wat_paired),
  clustering_distance_cols = as.dist(bray_wat_paired),
  labels_row = label_ab[ix_wat_paired],
  labels_col = label_ab[ix_wat_paired],
  main = "Bray-Curtis (Water only; paired sites)"
)

pheatmap(
  bray_sed_paired,
  clustering_distance_rows = as.dist(bray_sed_paired),
  clustering_distance_cols = as.dist(bray_sed_paired),
  labels_row = label_ab[ix_sed_paired],
  labels_col = label_ab[ix_sed_paired],
  main = "Bray-Curtis (Sediment only; paired sites)"
)

library(vegan)

ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]
NO3_ab <- as.numeric(sample_NO3_levels[ord])
PO4_ab <- as.numeric(sample_PO4_levels[ord])

habitat <- ifelse(grepl("\\bwater\\b", label_ab, ignore.case = TRUE), "Water",
  ifelse(grepl("\\bsediment\\b", label_ab, ignore.case = TRUE), "Sediment", "Other"))

eutro <- NO3_ab + PO4_ab

norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

```

```

bray <- vegdist(t(norm_seqtab), method = "bray")

adonis2(bray ~ eutro + habitat, permutations = 9999)

ix_wat <- which(habitat == "Water")

bray_wat <- vegdist(t(norm_seqtab)[ix_wat, , drop=FALSE], method = "bray")

adonis2(bray_wat ~ eutro[ix_wat], permutations = 9999)

ix_sed <- which(habitat == "Sediment")

bray_sed <- vegdist(t(norm_seqtab)[ix_sed, , drop=FALSE], method = "bray")

adonis2(bray_sed ~ eutro[ix_sed], permutations = 9999)

# --- ANOSIM (Bray-Curtis) for your dataset: Water vs Sediment (+ optional within-habitat eutro groups) ---

library(vegan)

# 0) Align metadata to seqtab columns
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- as.character(sample_label[ord])
NO3_ab <- as.numeric(sample_NO3_levels[ord])
PO4_ab <- as.numeric(sample_PO4_levels[ord])

# 1) Habitat from labels
habitat <- ifelse(grepl("\\bwater\\b", label_ab, ignore.case=TRUE), "Water",
  ifelse(grepl("\\bsediment\\b", label_ab, ignore.case=TRUE), "Sediment", "Other"))

# 2) Normalize + Bray-Curtis (as a MATRIX so we can subset)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))
bray_mat <- as.matrix(vegdist(t(norm_seqtab), method = "bray"))

# -----
# A) ANOSIM: Water vs Sediment
# -----
these <- which(habitat %in% c("Water", "Sediment"))
anosim_WS <- anosim(as.dist(bray_mat[these, these]),
  grouping = habitat[these],
  permutations = 9999)

anosim_WS
summary(anosim_WS)
plot(anosim_WS)

# -----
# B) OPTIONAL ANOSIM: eutrophication effect within WATER only (Low vs High)
# -----
eutro <- NO3_ab + PO4_ab
eutro_group <- factor(ifelse(eutro <= median(eutro, na.rm=TRUE), "Low", "High"),
  levels = c("Low", "High"))

ix_wat <- which(habitat == "Water" & !is.na(eutro))
anosim_wat_eutro <- anosim(as.dist(bray_mat[ix_wat, ix_wat]),
  grouping = eutro_group[ix_wat],
  permutations = 9999)

anosim_wat_eutro
summary(anosim_wat_eutro)
plot(anosim_wat_eutro)

# -----
# C) OPTIONAL ANOSIM: eutrophication effect within SEDIMENT only (Low vs High)
# -----
ix_sed <- which(habitat == "Sediment" & !is.na(eutro))
anosim_sed_eutro <- anosim(as.dist(bray_mat[ix_sed, ix_sed]),
  grouping = eutro_group[ix_sed],
  permutations = 9999)

anosim_sed_eutro
summary(anosim_sed_eutro)
plot(anosim_sed_eutro)

#----PERMANOVA---#

```

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library(vegan)

# Ensure metadata aligns with sample columns in seqtab
ord <- match(colnames(seqtab), course_id)
stopifnot(!any(is.na(ord)))

label_ab <- sample_label[ord]
NO3_ab <- as.numeric(sample_NO3_levels[ord])
PO4_ab <- as.numeric(sample_PO4_levels[ord])

# Define habitat groups
is_water <- grepl("\\bwater\\b", label_ab, ignore.case = TRUE)
is_sediment <- grepl("\\bsediment\\b", label_ab, ignore.case = TRUE)
habitat <- ifelse(is_water, "Water", ifelse(is_sediment, "Sediment", "Other"))

# Normalize to relative abundance per sample (recommended before Bray-Curtis)
norm_seqtab <- apply(seqtab, 2, function(x) x / sum(x))

# Bray-Curtis distances among ALL samples
bray <- vegdist(t(norm_seqtab), method = "bray")

# Eutrophication index
eutro <- NO3_ab + PO4_ab

# PERMANOVA: All samples (eutro + habitat)
adonis2(bray ~ eutro + habitat, permutations = 9999)

# PERMANOVA: Water only (eutro effect)
ix_wat <- which(habitat == "Water")
bray_wat <- vegdist(t(norm_seqtab)[ix_wat, , drop=FALSE], method = "bray")
adonis2(bray_wat ~ eutro[ix_wat], permutations = 9999)

# PERMANOVA: Sediment only (eutro effect)
ix_sed <- which(habitat == "Sediment")
bray_sed <- vegdist(t(norm_seqtab)[ix_sed, , drop=FALSE], method = "bray")
adonis2(bray_sed ~ eutro[ix_sed], permutations = 9999)

```